

NUTRIENTS

The nutrient is a substance that provides nourishment for growth and maintenance of life.

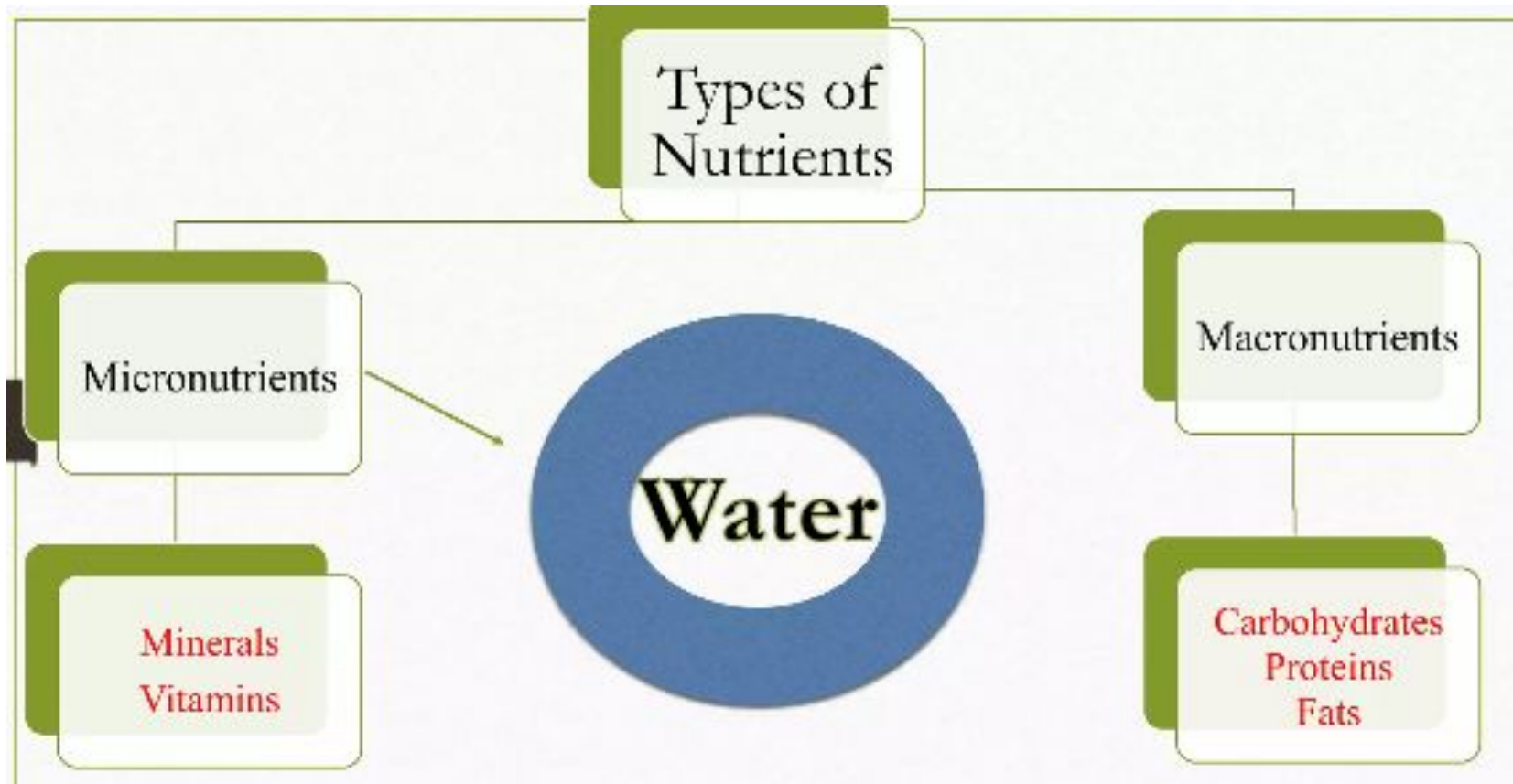
- Good nutrition helps fuel our body.
- Proper nutrition also helps protect you from illness and disease such as heart disease, diabetes, cancer, and osteoporosis.
- There are two major classes of nutrients in food:
macronutrients and micronutrients.

<https://pressbooks.bccampus.ca/humannutrition/chapter/digestion-and-absorption-of-carbohydrates/>

<https://pressbooks.library.vcu.edu/biol217vcu/chapter/6d-protein-digestion-absorption/>

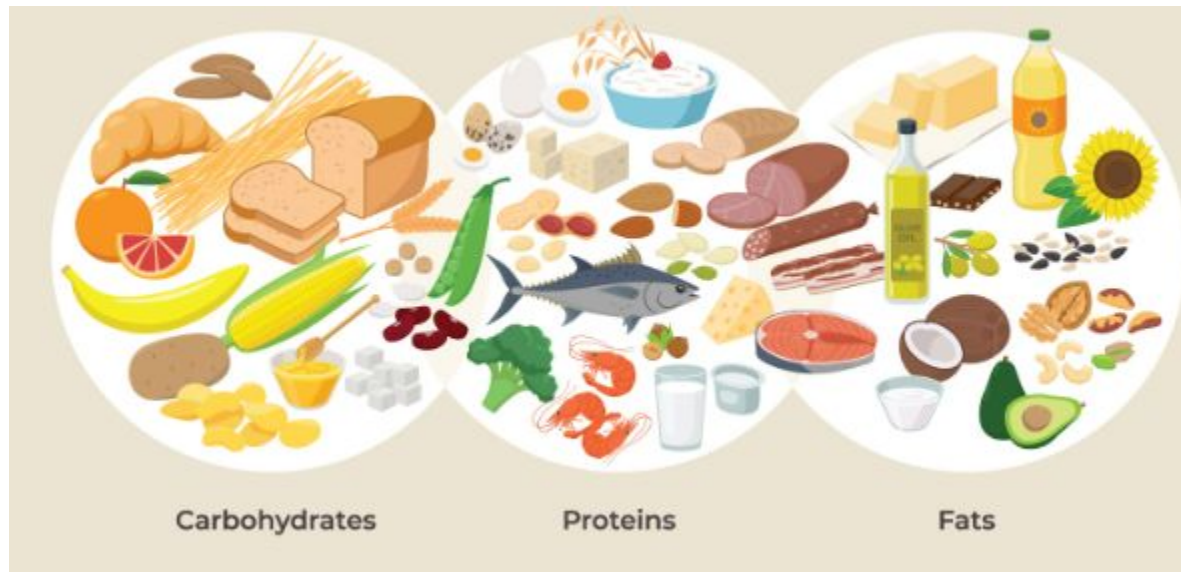
NUTRIENTS

Classification of Nutrients



MACRONUTRIENTS

- Macronutrients are a group of nutrients that provide your body with energy and the components it needs to maintain its structure and functions.
- Macronutrients include carbohydrates, protein, and fat. Your body needs them in relatively larger amounts than other nutrients, hence the term “macro.”



MACRONUTRIENTS

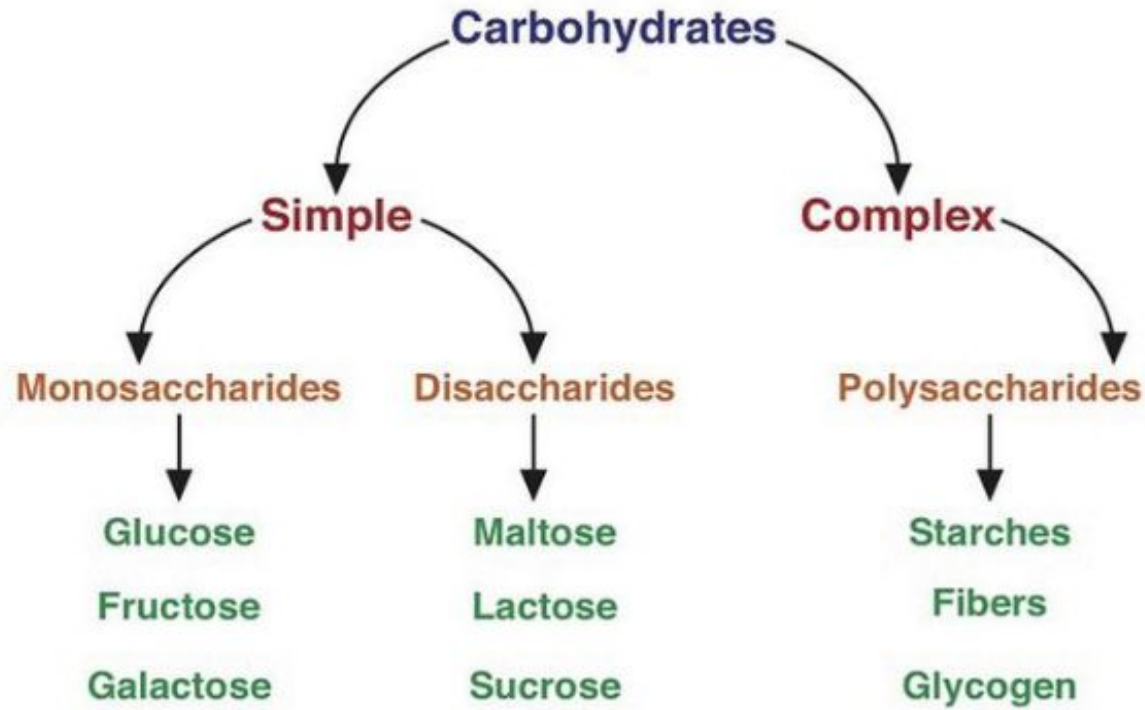
CARBOHYDRATES

- Carbohydrates are organic compounds made of carbon, hydrogen, and oxygen that serve as the main source of energy for the body. They are found in foods like rice, bread, fruits, vegetables, and sugar.
- Scientifically, carbohydrates include **sugars, starches, and cellulose**.
- Most carbohydrates follow the formula $C_n(H_2O)_n$, meaning they are hydrates of carbon.
- These are widely distributed molecules in both plant and animal tissues
- Plants are considerably more abundant in carbohydrates in comparison to animals.
- They also occur as food reserves in the storage organs of plants and the liver and muscles of animals. They are an essential source of energy required for the various metabolic activities of living organisms.

CARBOHYDRATES

Classification of Carbohydrates

Carbohydrates are classified into three different groups based on the degree of polymerization.



CARBOHYDRATES

MONOSACCHARIDES

Monosaccharides are the **simplest form of carbohydrates**, consisting of a single sugar unit. They **cannot be hydrolyzed** into smaller carbohydrate units.

-Most monosaccharides obey:

- The monosaccharides, often called $C_nH_{2n}O_n$ sugars, are compounds that possess a free aldehyde ($—CHO$) or ketone ($=CO$) group and two or more hydroxyl ($—OH$) groups.
- Monosaccharides are considered as fuel molecules that are involved in the formation of polymers like polysaccharides and nucleic acids.

CARBOHYDRATES

Classification of Monosaccharides

Monosaccharides are **further divided into** different groups based on various characteristics like the placement of the carbonyl group, the number of carbon atoms it contains, and the stereochemistry of the carbon atom.

1. Based on number of carbon atoms

- **Trioses (3C):** Glyceraldehyde
- **Tetroses (4C):** Erythrose
- **Pentoses (5C):** Ribose, Xylose
- **Hexoses (6C):** Glucose, Fructose, Galactose
- **Heptoses (7C):** Sedoheptulose

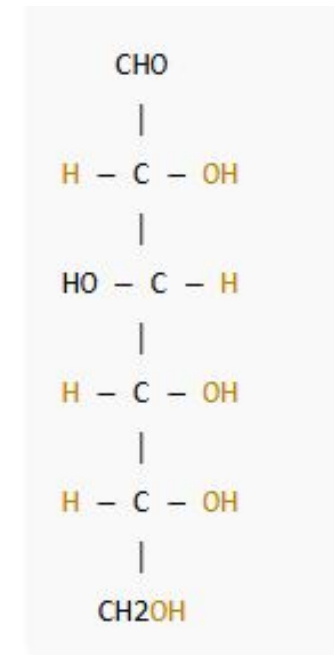
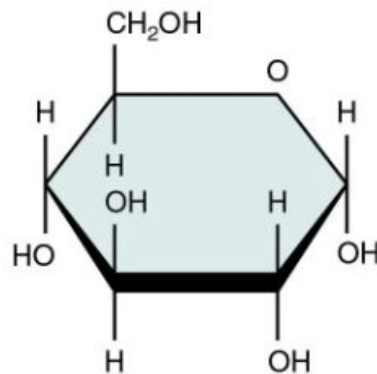
2. Based on functional group

- **Aldoses:** contain an aldehyde group (CHO)
Example: Glucose, Galactose
- **Ketoses:** contain a ketone group (C=O)
Example: Fructose

CARBOHYDRATES

Eg. of Monosaccharides-Glucose

- Glucose is an important monosaccharide that provides energy and structure to different parts of a cell.
- Glucose is a six-carbon compound with the molecular formula $C_6H_{12}O_6$. It is an aldohexose with six carbons and a free aldehyde group.
- The glucose molecule exists in both open-chain and ring form, the latter formed as a result of intermolecular interaction between the aldehyde carbon and the C-5 hydroxyl group.
- Glucose exists in two forms; α -glucose and β -glucose.
- The α -glucose forms the tasty part of the plants like fruits and flowers, whereas, the β -glucose forms the structurally hard part of the plant like the stem and roots.



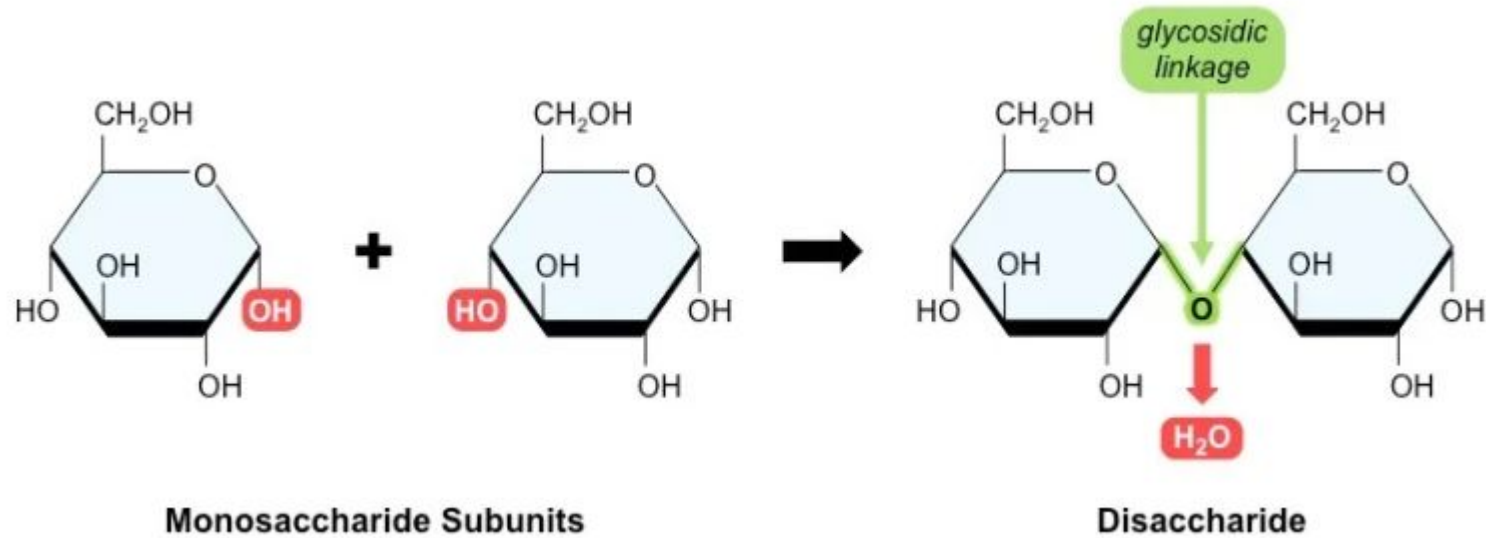
CARBOHYDRATES

DISACCHARIDES

- Glycosidic bonds are created when two monosaccharide molecules undergo a condensation process, forming disaccharides. They offer a rapid energy source and are frequently present in meals and drinks. Common examples of disaccharides are:
- **Sucrose (glucose + fructose):** Common table sugar composed of one molecule of glucose bonded to one molecule of fructose. It is found in sugarcane, sugar beets, and various sweetened foods.
- **Lactose (glucose + galactose):** Also known as milk sugar, it is composed of one molecule of glucose bonded to one molecule of galactose. It is found in milk and dairy products.
- **Maltose (glucose + glucose):** Formed during the breakdown of starch, maltose consists of two glucose molecules bonded together. It is found in germinating grains and malted foods.

CARBOHYDRATES

FORMATION OF DISACCHARIDES



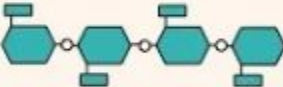
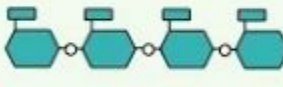
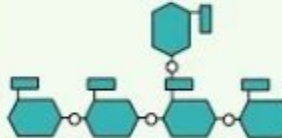
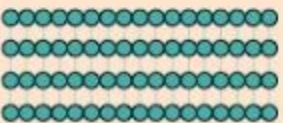
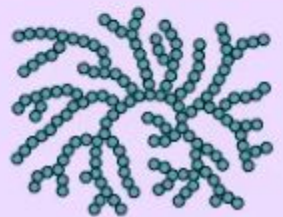
CARBOHYDRATES

POLYSACCHARIDES

- Polysaccharides are sugar molecules with more than ten monosaccharide units bonded together by glycosidic bonds.
- Polysaccharides are also termed glycan.
- Polysaccharides are a long chain of monosaccharides where the polysaccharide can be either homopolysaccharide or heteropolysaccharide.
- Homopolysaccharides are formed of identical monosaccharides, and heteropolysaccharides are formed of different monosaccharides.
- Polysaccharides have different forms depending on the monosaccharides present and the carbon atoms connected to one another.
- Some polysaccharides are linear, while others are branched.

CARBOHYDRATES

POLYSACCHARIDES

	Cellulose	Starch		Glycogen
		Amylose	Amylopectin	
Source	Plant	Plant	Plant	Animal
Subunit	β -glucose	α -glucose	α -glucose	α -glucose
Bonds	1-4	1-4	1-4 and 1-6	1-4 and 1-6
Branches	No	No	Yes (~per 20 subunits)	Yes (~per 10 subunits)
Diagram				
Shape				

CARBOHYDRATES

Examples of Polysaccharides

1. Starch

Starch is a polysaccharide comprising glucose monomers joined together by glycosidic linkages. Starch is an organic compound found in all living plants which is manufactured from the excess glucose produced during photosynthesis. Starch is the form of reserve food in plants stored in chloroplasts in the form of granules and storage organs like roots, tuber, stem, and seeds.

2. Glycogen

Glycogen is a branched polysaccharide that is a major form of glucose in animals and humans. It is often termed as 'animal starch' and is stored in the liver and muscles of animals.

SIMPLE AND COMPLEX CARBOHYDRATES

SIMPLE CARBS

- Simple carbs have a very basic chemical structure.
- They may be monosaccharides comprising a single sugar molecule, like glucose. Or they may be disaccharides, which have two simple sugars linked together, as with lactose (milk sugars).
- Enzymes in the small intestine break them down before they enter the bloodstream.
- Any sugar that isn't used right away is stored as fat. So eating foods with lots of added sugar can add to weight gain.



SIMPLE AND COMPLEX CARBOHYDRATES

COMPLEX CARBS

- Complex carbs are made of long, complex chains of sugar molecules. These are called oligosaccharides and polysaccharides.
- Complex carbs take longer to digest than simple carbs do.
- This means they have a less immediate impact on blood sugar, causing it to rise more slowly. They also fill you up for longer, which can help with portion control and weight management.

✓ **COMPLEX CARBS**



GOOD CARBS AND BAD CARBS

Good carbs are:

- Low or moderate in calories
- High in nutrients
- Devoid of refined sugars and refined grains
- High in naturally occurring fiber
- Low in sodium
- Low in saturated fat
- Very low in, or devoid of, cholesterol and trans fats

Bad carbs are:

- High in calories
- Low in many nutrients
- Full of refined sugars, like white sugar, honey and High in refined grains like white flour
- Low in fiber
- High in sodium
- Sometimes high in saturated fat
- Sometimes high in cholesterol and trans fats

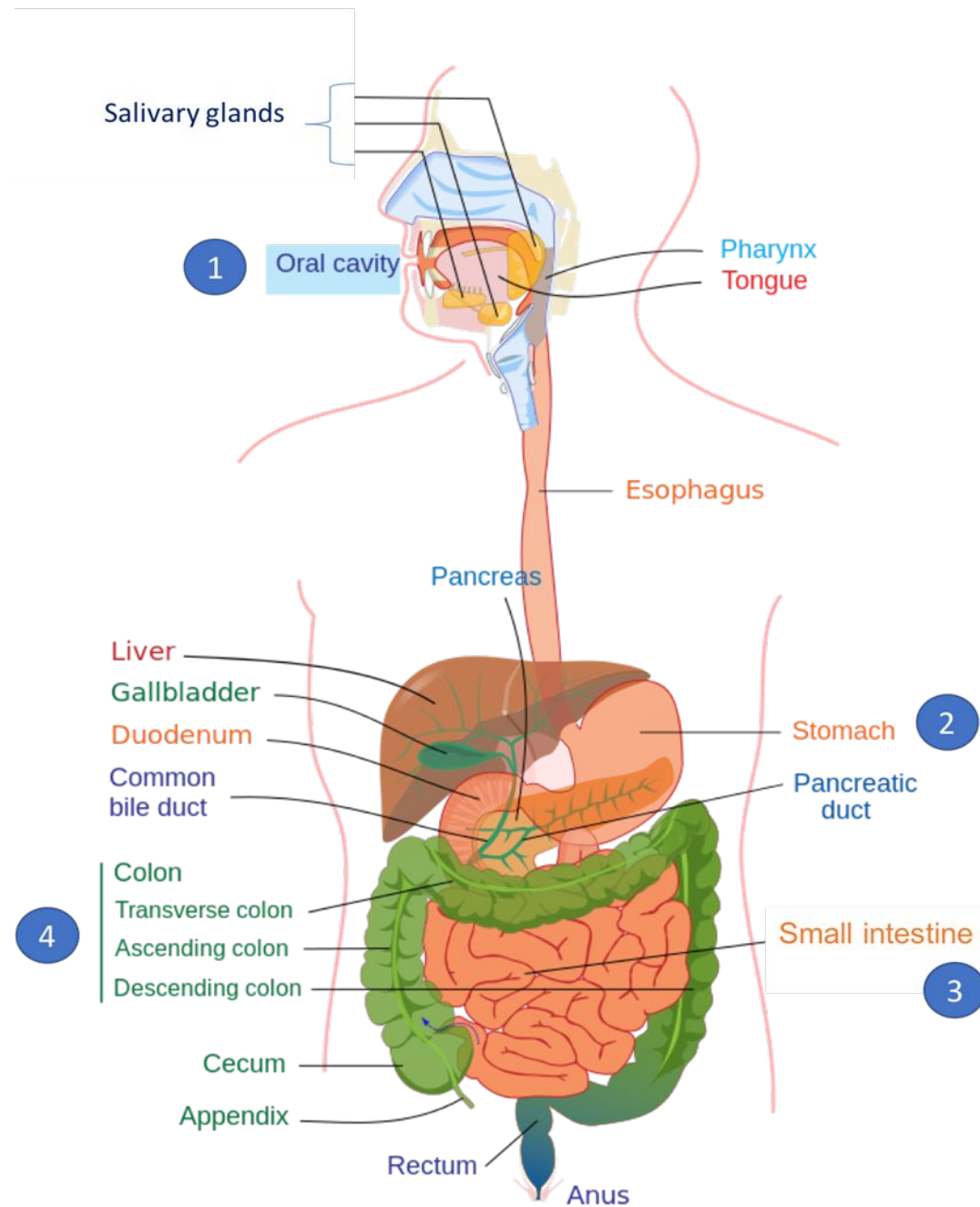
GLYCEMIC INDEX

Glycemic Index (GI) is a measure of **how quickly a carbohydrate-containing food raises blood glucose levels** after eating.

- The Glycemic Index ranks foods on a scale of **0 to 100** based on how fast they increase blood sugar compared to **pure glucose (GI = 100)**.
- Foods with high GI are digested quickly → cause a rapid rise in blood sugar.
- Foods with low GI are digested slowly → cause a gradual rise in blood sugar.
(one of the secrets to long term health).

GI Value	Meaning	Blood Sugar Effect	Examples
Below 55	Low GI	Slow increase	Apple, oats, milk, lentils
56–69	Medium GI	Moderate increase	Brown rice, banana
70 and above	High GI	Fast increase	White bread, potato, glucose

DIGESTIVE SYSTEM

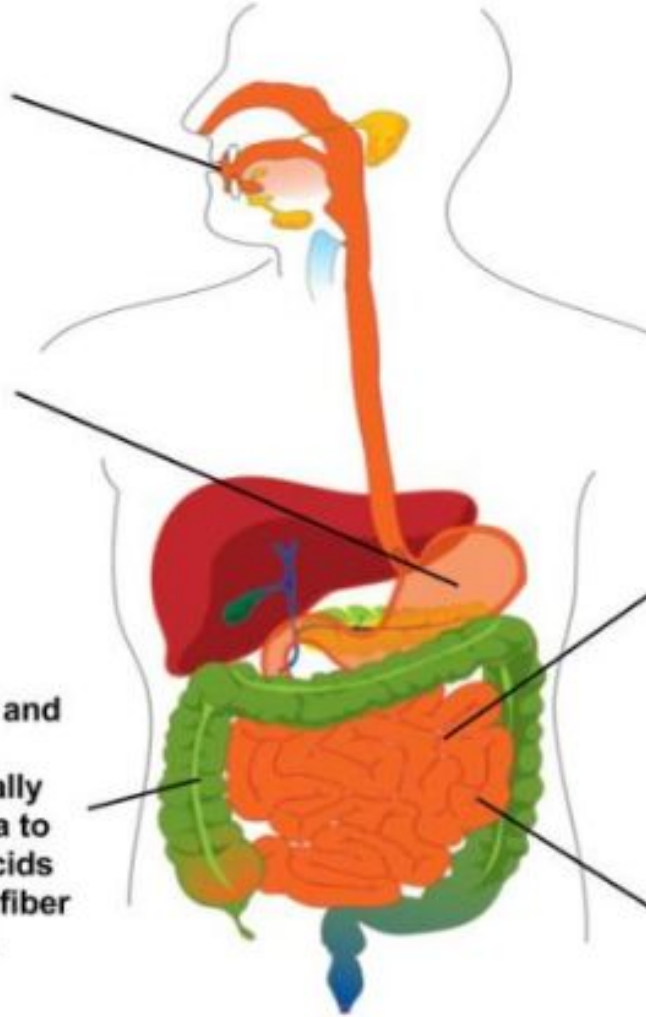


DIGESTION AND ABSORPTION OF CARBOHYDRATES

1. Mouth: The enzyme salivary amylase begins breaking down starch into shorter polysaccharides.

2. Stomach: Salivary amylase is inactivated and no further carbohydrate digestion occurs.

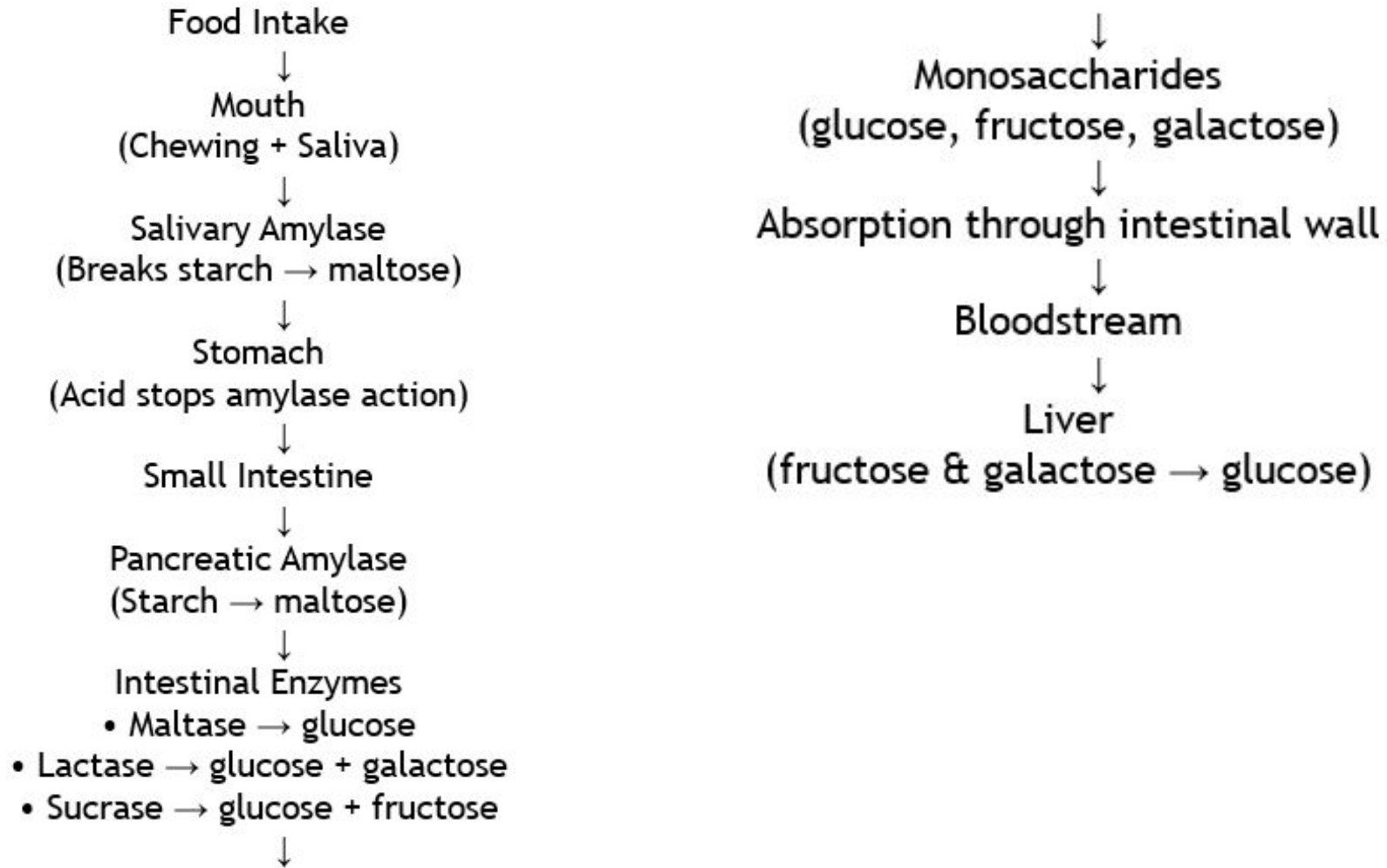
5. Large intestine: Fiber and other indigestible carbohydrates are partially broken down by bacteria to form short chain fatty acids and gas. The remaining fiber is excreted in the feces.



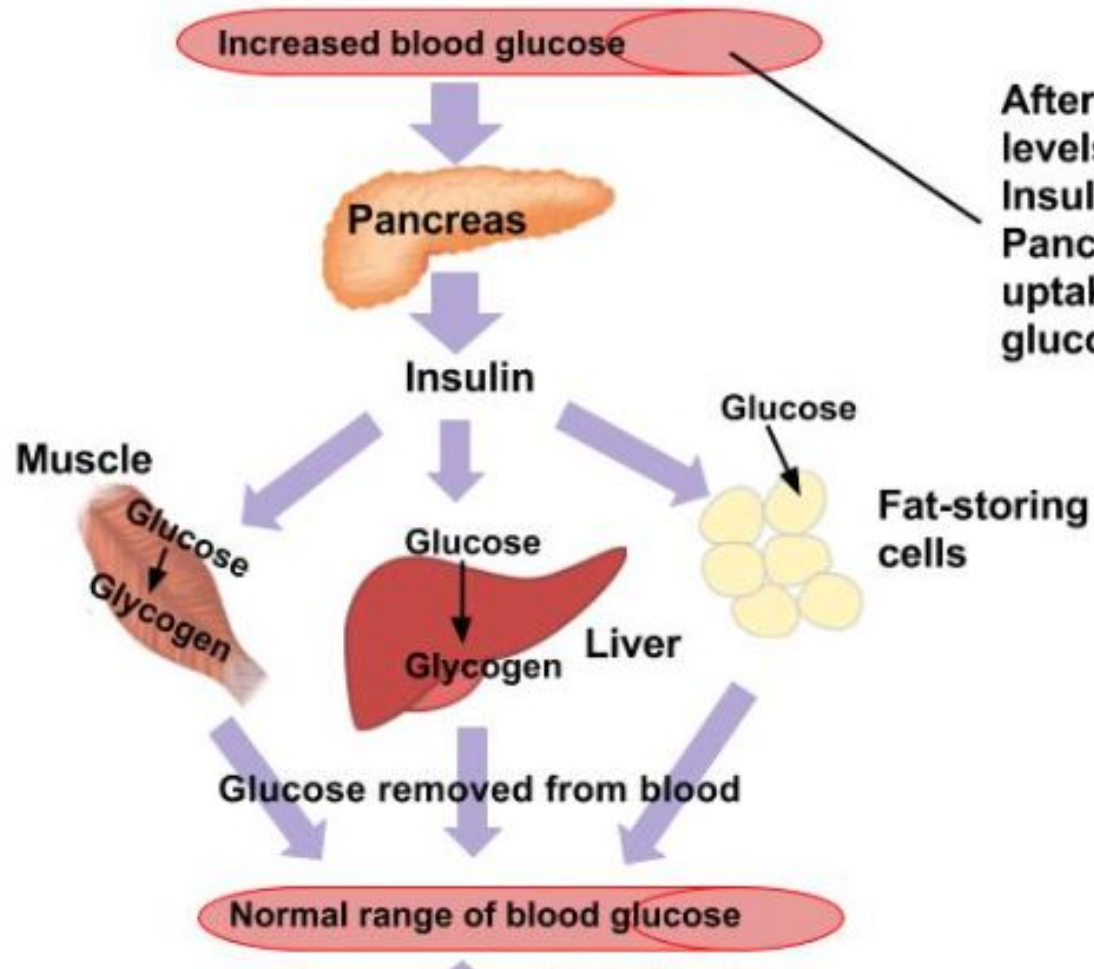
3. Small intestine: Majority of starch digestion and breakdown of disaccharides occur here. The enzyme pancreatic amylase breaks down starch into monosaccharides, disaccharides, and oligosaccharides.

4. The digestion of carbohydrates is completed by the enzymes attached to the brush border of the small intestinal villi. Here, the disaccharides and oligosaccharides are broken down into monosaccharides.

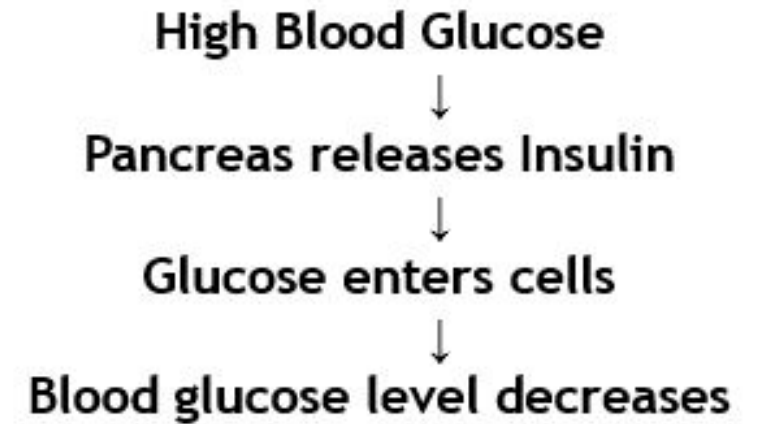
DIGESTION AND ABSORPTION OF CARBOHYDRATES



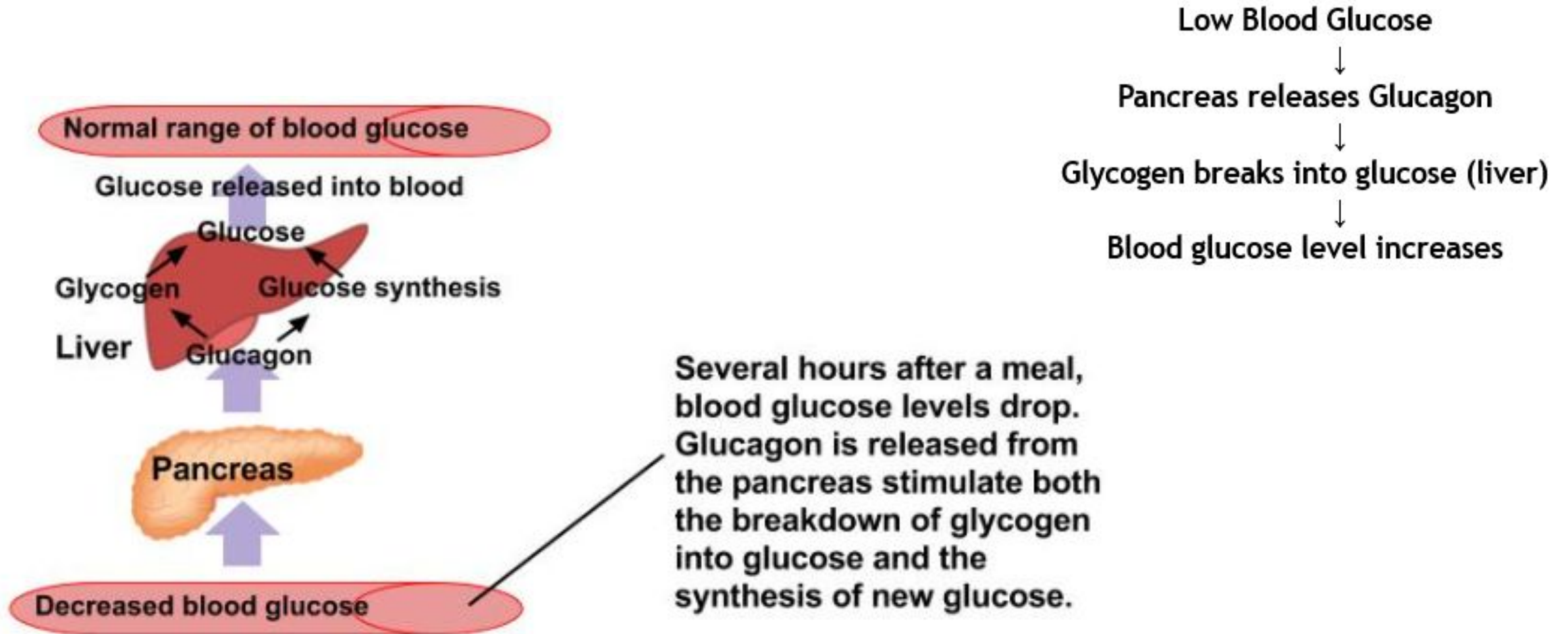
BLOOD GLUCOSE REGULATION



After a meal, blood glucose levels immediately increase. Insulin is released from the Pancreas, stimulating the uptake and storage of glucose.



BLOOD GLUCOSE REGULATION



RECOMMENDED DAILY ALLOWANCES

Recommended Dietary Allowances (RDA) for carbohydrates in India based on the ICMR-NIN (Indian Council of Medical Research – National Institute of Nutrition) guidelines is:

100–130 g per day for individuals aged 1 year and above — this ensures sufficient glucose supply for the brain and normal body functions.

FUNCTIONS OF CARBOHYDRATES

1) Main Source of Energy

- Provide 4 kcal per gram.
- Primary fuel for brain, nervous system, and red blood cells.
- Maintains normal physical and mental activity.

2) Protein Sparing Action

- When adequate carbohydrates are available, **proteins are spared** for their main role (growth and tissue repair) instead of being used for energy.

3) Fat Metabolism & Prevention of Ketosis

- Carbohydrates are necessary for complete oxidation of fats.
- Lack of carbs leads to incomplete fat breakdown → ketone body formation → ketosis.

4) Storage as Glycogen

- Excess glucose is stored as **glycogen** in liver and muscles.
- Acts as a readily available energy reserve during fasting or exercise.

FUNCTIONS OF CARBOHYDRATES

5) Structural and Functional Role

- Components of important biological molecules:
 - **Ribose** in RNA
 - **Deoxyribose** in DNA
 - Part of **glycoproteins** and **glycolipids** in cell membranes.

6) Regulation of Blood Glucose

- Maintain **normal blood sugar levels**.
- Provide continuous glucose supply between meals.

7) Dietary Fiber Function (Non-digestible carbohydrates)

- Improve **bowel movement** and prevent constipation.
- Reduce **cholesterol absorption**.
- Help in **blood sugar control**.
- Promote **gut health**.

FUNCTIONS OF CARBOHYDRATES

8) Role in Taste and Satiety

- Provide sweetness and palatability to food.
- Give a feeling of fullness (satiety).

9) Detoxification Function

- Help the liver in detoxification by forming **glucuronic acid**, which combines with toxins and drugs for excretion.

10) Prevention of Muscle Breakdown

- During starvation or low carb intake, the body uses muscle protein for glucose. Adequate carbs **prevent muscle wasting**.

SOURCES OF CARBOHYDRATES

Carbohydrates are mainly obtained from **plant-based foods** and form the major part of a typical diet.

1.Cereals and Millets (Richest sources)

- Rice, wheat, maize (corn)
- Ragi, jowar, bajra, oats, barley

2) Pulses and Legumes

- Bengal gram, green gram, black gram
- Lentils, peas, beans, chickpeas, rajma

3) Roots and Tubers

- Potato, sweet potato
- Tapioca, yam

4) Fruits

- Banana, apple, mango, grapes, orange
- Dates, figs, pineapple
- Contain natural sugars: **fructose and glucose**

SOURCES OF CARBOHYDRATES

5) Vegetables

- Carrot, beetroot, corn
- Pumpkin, peas

6) Milk and Milk Products

- Milk, curd, yogurt
- Contain **lactose** (milk sugar)

7) Sugars and Sweeteners

- Table sugar, jaggery, honey
- Glucose powder, syrups

8) Processed and Bakery Foods

- Biscuits, cakes, pastries
- Soft drinks, chocolates, sweets

DEFICIENCY OF CARBOHYDRATES

Carbohydrate deficiency occurs when the body **does not receive enough carbohydrates** to meet its energy needs.

- **1) Low Energy and Fatigue**

- Carbohydrates are the main energy source.
- Deficiency leads to weakness, tiredness, and reduced work capacity.

- **2) Hypoglycemia (Low Blood Sugar)**

- Inadequate glucose supply causes:
 - Dizziness, Headache, Confusion, Fainting in severe cases.

- **3) Ketosis**

- In the absence of carbohydrates, the body burns fat for energy.
- Leads to production of ketone bodies → ketosis (nausea, bad breath, dehydration).

- **4) Muscle Wasting**

- Body starts breaking down muscle protein to produce glucose.
- Results in loss of muscle mass.

DEFICIENCY OF CARBOHYDRATES

5) Poor Brain Function

- Brain depends mainly on glucose.
- Deficiency affects concentration, memory, and alertness.

6) Constipation

- Low intake of carbohydrate-rich foods often means low fiber intake.
- Leads to **poor bowel movement**.

7) Weight Loss

- Due to breakdown of fat and muscle for energy.

8) Nutritional Imbalance

- Very low carbohydrate diets may lack essential vitamins, minerals, and fiber found in carb-rich foods.